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Fluid phase equilibria of binary and ternary mixtures of supercritical carbon dioxide with low-volatility organic substances up to 100 MPa and 393 K: cosolvency effects and miscibility windows

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ABSTRACT

Kordikowski, A. and Schneider, G.M., 1993. Fluid phase equilibria of binary and ternary mixtures of supercritical carbon dioxide with low-volatility organic substances up to 100 MPa and 393 K: Cosolvency effects and miscibility windows. *Fluid Phase Equilibria* 90: 149–162.

Isothermal $p(w)$ diagrams at 353.2 and 393.2 K as well as isobaric $T(w)$ diagrams at 25 MPa were determined for binary mixtures of CO_2 with 1-dodecanol and the *n*-alkanes hexadecane nonadecane, eicosane and tetracosane. In addition measurements were taken for the ternary systems CO_2 + *n*-alkane + 1-dodecanol (where *n*-alkane = hexadecane, nonadecane, eicosane or tetracosane) under the same conditions of temperature and pressure. For the system CO_2 + nonadecane + 1-dodecanol additional experiments were carried out at 308.2 K. The resulting ternary critical curves exhibit 'cosolvency effects' and 'miscibility windows.'

INTRODUCTION

The understanding of the solubility of low-volatility organic substances is of great interest, especially for applications in supercritical fluid extraction. Compressed carbon dioxide is a good solvent for many organic substances (Schneider et al., 1980). Another highly volatile component is frequently added as a moderator (Brunner, 1983; Katzenski-Ohling and Schneider, 1987; Hölscher et al., 1989) in order to decrease or increase the solvent power of the fluid phase. Addition of another low-volatility component, however, is less common and produces interesting phase equilibria of the resulting ternary systems.

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EXPERIMENTAL

The experimental data were obtained by using an analytical high-pressure cell. A detailed description of the apparatus is given in earlier papers by Konrad (1982), Konrad et al. (1983) and Deiters and Schneider (1986).

The measuring technique is described briefly in the following. The low-volatility components were placed in the pressure vessel. The pressure was increased either with a hand pump for liquid substances or with a compressor for gaseous components. The pressure was measured by a Heise gauge within ± 0.1 MPa. The temperature was measured to within ± 0.1 K by a thermocouple placed in the fluid mixture.

The p, T, w' and w'' (w = mass fraction: $'$, $''$ = coexisting phases) equilibrium data were determined by sampling and subsequent analysis. During sampling the sample splits into two phases in the evacuated analysis system: a gaseous phase containing practically all CO_2 and a condensed phase containing the low-volatility components. The mass of the gaseous component (CO_2) could be calculated from an equation of state. The low-volatility components were dissolved and rinsed out of the sampling system by means of an organic solvent. After addition of an external standard they were analysed by gas chromatography.

The purity of the alkanes and of 1-dodecanol was better than 99% respectively (Fluka AG). All substances were used without further purification.

RESULTS

In this paper data for five binary and four ternary systems are presented. The measurements for the binary systems were performed at 353.2 K, at 393.2 K for the isothermal measurements and at 25 MPa for the isobaric measurements. For the ternary systems the same temperatures and pressures were used. For the ternary system CO_2 + nonadecane + 1-dodecanol, additional measurements at 308.2 K were performed. All the experiments were carried out at pressures between 10 and 100 MPa and temperatures between 298.2 and 393.2 K.

Selected results for the binary and ternary systems are presented in Figs. 1–6 and in Tables 1 and 2 respectively. Sometimes more data are given in the Figures than in the Tables. For additional numerical data see Kordikowski (1992).

Carbon dioxide + hexadecane

Preceding measurements on this system have been extended to a temperature of 353.2 K in the present paper (see Table 1). The system has already been discussed elsewhere (see e.g. Schneider, 1991).

TABLE 1

Fluid–fluid phase equilibrium data for binary mixtures of CO_2 with 1-dodecanol, hexadecane, nonadecane, eicosane and tetracosane (w = mass fraction)

p (MPa)	w_1^g	w_1^l	w_2^g	w_2^l
CO_2 + 1-dodecanol (353.2 K)				
10.0	0.990	0.169	0.010	0.831
15.0	0.987	0.282	0.013	0.718
20.0	0.931	0.386	0.069	0.614
23.0	0.868	0.447	0.132	0.553
25.1	0.805	0.578	0.195	0.422
25.2	0.690	0.691	0.310	0.309
CO_2 + hexadecane (353.2 K)				
10.0	0.997	0.290	0.003	0.771
12.5	0.995	0.296	0.005	0.704
15.0	0.981	0.387	0.019	0.613
17.5	0.942	0.473	0.058	0.527
18.5	0.908	0.494	0.092	0.506
19.6	0.852	0.562	0.148	0.438
20.0	0.740	0.740	0.260	0.260
CO_2 + nonadecane (353.2 K)				
10.0	0.998	0.159	0.002	0.841
15.0	0.995	0.165	0.005	0.835
20.0	0.975	0.278	0.025	0.772
27.5	0.649	0.651	0.351	0.349

p (MPa)	w_1^g	w_1^l	w_2^g	w_2^l
CO_2 + nonadecane (393.2 K)				
10.0	0.999	0.126	0.001	0.874
15.0	0.995	0.165	0.005	0.835
20.0	0.986	0.244	0.014	0.756
25.0	0.955	0.361	0.045	0.639
27.5	0.914	0.408	0.086	0.592
30.0	0.786	0.502	0.214	0.498
30.3	0.688	0.691	0.312	0.319
CO_2 + nonadecane (25 MPa)				
T/K	w_1^g	w_1^l	w_2^g	w_2^l
303.15	0.336	0.856	0.664	0.144
313.15	0.412	0.840	0.588	0.160
323.15	0.472	0.812	0.528	0.188
333.15	0.815	0.487	0.185	0.513
343.15	0.808	0.470	0.192	0.530
353.15	0.866	0.428	0.134	0.572
363.15	0.893	0.415	0.107	0.485
373.15	0.911	0.378	0.089	0.622
383.15	0.930	0.355	0.070	0.645
CO_2 + eicosane (353.2 K)				
p (MPa)	w_1^g	w_1^l	w_2^g	w_2^l
10.0	0.998	0.143	0.002	0.857
15.0	0.992	0.189	0.008	0.811
20.0	0.968	0.275	0.032	0.725
25.0	0.882	0.363	0.118	0.637
27.5	0.856	0.447	0.144	0.553
30.0	0.651	0.550	0.349	0.450
30.2	0.630	0.630	0.370	0.370

p (MPa)	w_1^g	w_1^l	w_2^g	w_2^l
CO_2 + eicosane (393.2 K)				
10.0	0.999	0.119	0.001	0.881
15.0	0.997	0.147	0.003	0.853
20.0	0.980	0.200	0.020	0.800
25.0	0.964	0.272	0.036	0.728
27.5	0.942	0.393	0.058	0.607
30.0	0.894	0.416	0.106	0.584
31.0	0.858	0.499	0.142	0.501
32.0	0.688	0.691	0.312	0.309
CO_2 + tetracosane (353.2 K)				
10.0	0.997	0.126	0.003	0.874
15.0	0.994	0.178	0.006	0.822
20.0	0.986	0.225	0.014	0.775
25.0	0.965	0.252	0.035	0.748
30.0	0.919	0.303	0.081	0.697
32.5	0.896	0.328	0.104	0.672
35.0	0.866	0.337	0.134	0.663
37.5	0.834	0.397	0.166	0.603
40.0	0.827	0.450	0.173	0.550
47.5	0.805	0.468	0.195	0.532
CO_2 + tetracosane (393.2 K)				
10.0	0.999	0.097	0.001	0.903
15.0	0.998	0.144	0.002	0.856
20.0	0.995	0.171	0.005	0.829
25.0	0.984	0.228	0.016	0.772
30.0	0.962	0.306	0.038	0.694
5.0	0.922	0.366	0.078	0.634
7.5	0.878	0.470	0.122	0.530
9.8	0.654	0.653	0.346	0.347

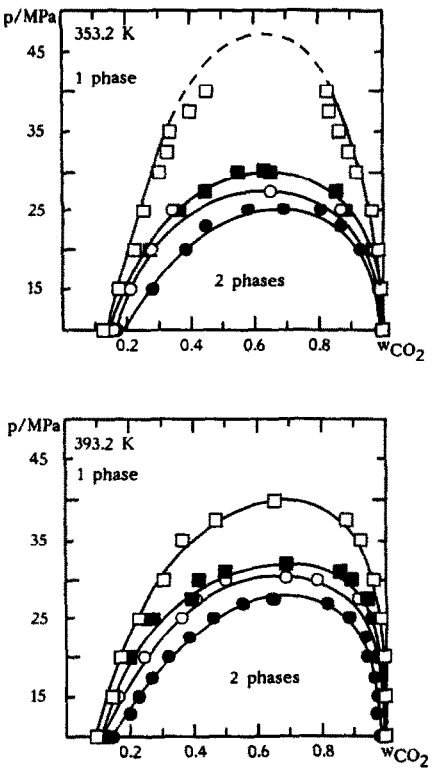


Fig. 1. $p(w)$ diagrams at 353.2 K a) and 393.2 K b) for the system CO_2 + 1-dodecanol (393.2 K; Schneider, 1991) (●), CO_2 + nonadecane (○), CO_2 + eicosane (■) and CO_2 + tetracosane (□).

Carbon dioxide + 1-dodecanol

The isothermal $p(w)$ phase diagram of the system carbon dioxide + 1-dodecanol at 353.2 K is shown in Fig. 1a. The mutual miscibility increases with increasing pressure. The critical pressure at 353.2 K is $p_c = 25.2$ MPa and the critical mass fraction of carbon dioxide is $w_c = 0.69$. This binary system corresponds to type III according to the classification of van Konynenburg and Scott with a continuous critical line from a liquid–liquid to a liquid–gas equilibrium. Additional measurements for this system have already been published by Spee and Schneider (1991) and Schneider (1991).

Carbon dioxide + nonadecane

The isothermal $p(w)$ phase diagrams of this system at 353.2 and 393.2 K are shown in Figs. 1a and 1b. This system is very similar to that of CO_2 + 1-dodecanol. The critical pressure at 353.2 K is $p_c = 27.5$ MPa with a

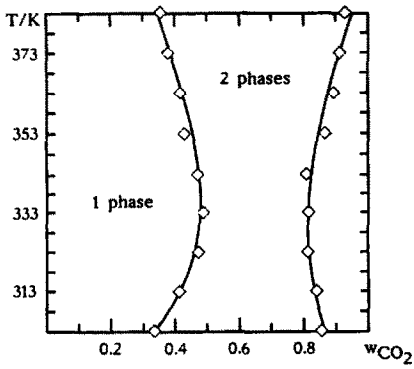


Fig. 2. $T(w)$ diagram of the system CO_2 + nonadecane at 25 MPa.

critical mass fraction of carbon dioxide of $w_c = 0.65$. At 393.2 K the critical pressure increased to 30.3 MPa with a corresponding mass fraction of $w_c = 0.69$.

The isobaric $T(w)$ phase diagram of this system (Fig. 2) at 25 MPa shows a non-interrupted heterogeneous region across the whole temperature range. At temperatures below 333.2 K barotropy occurs.

Carbon dioxide + eicosane

The system CO_2 + eicosane was also investigated at 353.2 and 393.2 K. As shown in Figs. 1a and 1b the binodal curves in the isothermal $p(w)$ diagram are very similar to those of the other systems. The critical pressure at 353.2 K is $p_c = 30.2$ MPa with a corresponding mass fraction of $w_c = 0.63$. At 393.2 K the values have changed to $p_c = 32.0$ MPa and $w_c = 0.69$.

Carbon dioxide + tetracosane

This system was investigated at the same temperatures as the other systems. The shape of the binodal curve is again similar to the other systems. At 393.2 K (Fig. 1b) the binodal curve exhibits a critical pressure of $p_c = 39.8$ MPa with a critical mass fraction of carbon dioxide of $w_c = 0.68$. At 353.2 K (Fig. 1a) the critical pressure has considerably increased. At this temperature a more liquid–liquid like equilibrium is found in contrast to a more gas–liquid like equilibrium at the higher temperature as in the other systems.

Carbon dioxide + hexadecane + 1-dodecanol

Selected data are presented in Table 2 for 353.2 K. They have already been discussed elsewhere (Spee, 1990; Schneider, 1991).

TABLE 2

Fluid–fluid phase equilibrium data for selected ternary CO_2 + n-alkane + 1-dodecanol systems (w = mass fraction)

p (MPa)	T (K)	w_1^g	w_1^l	w_2^g	w_2^l	w_3^g	w_3^l
CO_2 + hexadecane + 1-dodecanol (353.2 K)							
10.0		0.992	0.192	0.003	0.216	0.005	0.592
10.0		0.995	0.220	0.004	0.577	0.001	0.203
12.5		9.991	0.275	0.003	0.192	0.006	0.533
12.5		0.991	0.296	0.006	0.520	0.003	0.184
15.0		0.977	0.348	0.013	0.319	0.010	0.333
15.0		0.978	0.335	0.008	0.176	0.014	0.489
17.5		0.942	0.408	0.021	0.156	0.037	0.436
17.5		0.933	0.460	0.047	0.400	0.020	0.140
18.5		0.856	0.527	0.104	0.350	0.040	0.123
19.0		0.570	0.575	0.319	0.314	0.111	0.111
20.0		0.702	0.604	0.146	0.186	0.152	0.210
20.1		0.650	0.651	0.167	0.167	0.183	0.182
20.5		0.820	0.532	0.056	0.123	0.124	0.345
22.3		0.702	0.702	0.065	0.065	0.233	0.233
CO_2 + nonadecane + 1-dodecanol (308.2 K)							
20.0		0.364	0.867	0.310	0.063	0.326	0.070
20.0		0.408	0.813	0.405	0.123	0.187	0.064
30.0		0.430	0.829	0.270	0.082	0.300	0.089
30.0		0.505	0.704	0.277	0.163	0.218	0.132
40.0		0.400	0.802	0.141	0.058	0.459	0.140
40.0		0.484	0.743	0.210	0.116	0.306	0.141
60.0		0.444	0.818	0.133	0.051	0.423	0.131
60.0		0.504	0.820	0.458	0.157	0.038	0.023
80.0		0.432	0.804	0.134	0.053	0.434	0.143
80.0		0.485	0.806	0.475	0.148	0.040	0.026
95.0		0.551	0.616	0.352	0.281	0.097	0.103
CO_2 + eicosane + 1-dodecanol (393.2 K)							
10.0		0.985	0.132	0.006	0.419	0.009	0.449
10.0		0.992	0.126	0.001	0.285	0.007	0.589
15.0		0.964	0.190	0.015	0.397	0.021	0.413
15.0		0.981	0.177	0.002	0.253	0.017	0.570
20.0		0.958	0.248	0.006	0.222	0.036	0.530
25.0		0.887	0.320	0.046	0.350	0.067	0.330
25.0		0.885	0.359	0.019	0.127	0.096	0.514
26.0		0.862	0.324	0.055	0.319	0.083	0.357
26.0		0.854	0.454	0.011	0.051	0.135	0.495
26.9		0.642	0.642	0.071	0.072	0.287	0.286
26.9		0.647	0.647	0.109	0.108	0.244	0.246
27.0		0.568	0.537	0.037	0.041	0.395	0.422
CO_2 + eicosane + 1-dodecanol (25 MPa)							
	303.15	0.348	0.793	0.204	0.070	0.448	0.137
	303.15	0.341	0.831	0.111	0.036	0.548	0.133
	313.15	0.417	0.667	0.182	0.108	0.401	0.225
	313.15	0.389	0.794	0.106	0.036	0.505	0.170
	323.15	0.777	0.461	0.023	0.044	0.200	0.495
	323.15	0.739	0.569	0.046	0.081	0.215	0.350
	333.15	0.777	0.560	0.023	0.039	0.200	0.401
	333.15	0.706	0.626	0.051	0.072	0.243	0.302
	343.15	0.759	0.564	0.025	0.041	0.216	0.395
	353.15	0.866	0.488	0.113	0.463	0.021	0.049
	363.15	0.838	0.490	0.112	0.401	0.050	0.109
	363.15	0.639	0.578	0.066	0.079	0.295	0.343
	373.15	0.728	0.558	0.072	0.140	0.200	0.302

95.0	0.475	0.770	0.217	0.100	0.308	0.130
100.0	0.527	0.755	0.227	0.118	0.246	0.127
100.0	0.502	0.733	0.286	0.151	0.212	0.116
CO_2 + nonadecane + 1-dodecanol (353.2 K)						
10.0	0.998	0.159	0.001	0.436	0.001	0.405
10.0	0.994	0.154	0.003	0.683	0.003	0.163
15.0	0.986	0.214	0.005	0.414	0.009	0.372
15.0	0.982	0.209	0.006	0.283	0.012	0.508
20.0	0.905	0.336	0.044	0.340	0.051	0.324
20.0	0.926	0.317	0.051	0.556	0.023	0.127
22.5	0.841	0.415	0.038	0.125	0.121	0.460
22.5	0.847	0.447	0.114	0.450	0.039	0.103
23.1	0.631	0.631	0.241	0.242	0.128	0.127
23.9	0.628	0.628	0.081	0.082	0.291	0.290
CO_2 + nonadecane + 1-dodecanol (393.2 K)						
10.0	0.997	0.116	0.001	0.139	0.002	0.745
10.0	0.991	0.121	0.004	0.320	0.005	0.559
15.0	0.982	0.165	0.003	0.126	0.015	0.709
15.0	0.986	0.169	0.003	0.308	0.011	0.523
20.0	0.960	0.245	0.005	0.114	0.035	0.641
20.0	0.963	0.265	0.008	0.247	0.029	0.488
25.0	0.868	0.374	0.019	0.094	0.113	0.532
25.0	0.863	0.384	0.041	0.218	0.096	0.398
26.4	0.624	0.625	0.130	0.129	0.246	0.246
26.5	0.616	0.504	0.201	0.267	0.183	0.229
26.6	0.560	0.560	0.234	0.234	0.206	0.206
26.7	0.533	0.536	0.072	0.071	0.395	0.393
27.6	0.650	0.650	0.260	0.258	0.090	0.092
CO_2 + nonadecane + 1-dodecanol (25 MPa)						
303.15	0.501	0.812	0.416	0.145	0.083	0.043
303.15	0.332	0.850	0.132	0.037	0.536	0.113
373.15	0.774	0.506	0.049	0.098	0.207	0.396
383.15	0.841	0.446	0.048	0.196	0.111	0.358
383.15	0.853	0.468	0.062	0.267	0.085	0.265
393.15	0.876	0.414	0.022	0.119	0.102	0.467
393.15	0.887	0.415	0.046	0.302	0.067	0.283
CO_2 + tetracosane + 1-dodecanol (353.2 K)						
10.0	0.995	0.145	0.001	0.082	0.004	0.773
10.0	0.997	0.133	0.001	0.223	0.002	0.644
20.0	0.931	0.255	0.012	0.192	0.057	0.553
20.0	0.939	0.226	0.018	0.406	0.043	0.367
25.0	0.780	0.452	0.018	0.053	0.202	0.495
25.0	0.760	0.461	0.047	0.117	0.193	0.422
25.5	0.633	0.632	0.039	0.039	0.328	0.329
26.7	0.625	0.625	0.182	0.183	0.193	0.192
27.5	0.865	0.395	0.086	0.491	0.049	0.114
30.0	0.829	0.446	0.116	0.449	0.055	0.105
33.2	0.654	0.654	0.274	0.274	0.072	0.072
CO_2 + tetracosane + 1-dodecanol (393.2 K)						
10.0	0.996	0.116	0.002	0.108	0.002	0.776
10.0	0.997	0.103	0.001	0.613	0.002	0.284
15.0	0.985	0.169	0.003	0.091	0.012	0.740
15.0	0.986	0.188	0.001	0.193	0.013	0.619
20.0	0.965	0.240	0.002	0.184	0.033	0.576
20.0	0.978	0.204	0.006	0.546	0.016	0.250
25.0	0.897	0.330	0.006	0.073	0.097	0.597
25.0	0.903	0.436	0.015	0.147	0.082	0.417
27.0	0.802	0.468	0.014	0.054	0.184	0.478
27.0	0.827	0.447	0.034	0.151	0.139	0.402
27.5	0.635	0.634	0.034	0.034	0.331	0.332
28.0	0.637	0.635	0.093	0.093	0.270	0.272
30.0	0.838	0.342	0.084	0.453	0.078	0.205

TABLE 2 (continued)

p (MPa)	T (K)	w_1^g	w_1^l	w_2^g	w_2^l	w_3^g	w_3^l
CO_2 + nonadecane + 1-dodecanol (25 MPa) (continued)							
	313.15	0.525	0.655	0.394	0.279	0.081	0.066
	313.15	0.361	0.881	0.063	0.017	0.576	0.102
	323.15	0.789	0.506	0.050	0.108	0.161	0.386
	323.15	0.598	0.598	0.368	0.368	0.034	0.034
	333.15	0.783	0.519	0.026	0.053	0.191	0.428
	343.15	0.741	0.528	0.029	0.051	0.230	0.421
	363.15	0.828	0.469	0.151	0.486	0.021	0.045
	373.15	0.814	0.486	0.021	0.058	0.165	0.456
	373.15	0.889	0.401	0.095	0.548	0.016	0.051
	383.15	0.837	0.458	0.076	0.273	0.087	0.269
	383.15	0.858	0.424	0.018	0.070	0.124	0.506
CO_2 + eicosane + 1-dodecanol (353.2 K)							
10.0		0.993	0.153	0.003	0.187	0.004	0.660
10.0		0.996	0.151	0.001	0.082	0.003	0.767
15.0		0.983	0.201	0.004	0.277	0.013	0.522
15.0		0.983	0.215	0.002	0.074	0.015	0.711
20.0		0.921	0.295	0.024	0.242	0.055	0.463
20.0		0.929	0.294	0.008	0.068	0.063	0.638
22.5		0.828	0.426	0.057	0.201	0.115	0.373
22.5		0.882	0.399	0.013	0.058	0.105	0.543
23.3		0.646	0.644	0.181	0.182	0.173	0.174
23.5		0.627	0.627	0.129	0.128	0.244	0.245
24.0		0.852	0.478	0.105	0.426	0.043	0.096
24.0		0.817	0.456	0.019	0.052	0.164	0.492
24.1		0.631	0.631	0.079	0.079	0.289	0.289
24.9		0.637	0.635	0.034	0.035	0.329	0.330
25.3		0.666	0.665	0.266	0.265	0.068	0.070
CO_2 + tetracosane + 1-dodecanol (393.2 K) (continued)							
31.5		0.738	0.509	0.164	0.338	0.098	0.153
31.7		0.624	0.624	0.251	0.250	0.125	0.126
CO_2 + tetracosane + 1-dodecanol (25 MPa)							
	323.15	0.809	0.435	0.096	0.325	0.095	0.240
	323.15	0.832	0.488	0.037	0.126	0.131	0.386
	333.15	0.881	0.439	0.090	0.327	0.099	0.234
	333.15	0.823	0.509	0.019	0.058	0.158	0.433
	343.15	0.788	0.449	0.104	0.324	0.108	0.227
	343.15	0.790	0.508	0.023	0.059	0.187	0.433
	353.15	0.834	0.425	0.073	0.336	0.093	0.239
	353.15	0.802	0.486	0.021	0.063	0.177	0.451
	363.15	0.856	0.353	0.058	0.400	0.086	0.247
	363.15	0.820	0.437	0.019	0.071	0.161	0.492

Carbon dioxide + nonadecane + 1-dodecanol

This ternary system was investigated isothermally at 353.2 and 393.2 K. As shown in Figs 4a and 4b the data are plotted in an isothermal phase prism. The front side and the left side of the prism represent the corresponding binary systems. The critical pressures of the binary systems are connected by a ternary critical line, which is shown in bold type. A projection of the ternary critical line against a carbon dioxide free mass fraction is given in Figs 5a and 5b. You can see a distinct pressure minimum on the curves corresponding to a so-called 'cosolvency effect.'

This system was also investigated under isobaric conditions at 25 MPa just below the pressure minimum of both binary critical lines (Fig. 3). The results are plotted in an isobaric phase prism shown in Fig. 4d. Here the ternary critical curve forms a closed loop. Inside this loop a homogeneous region exists. This effect is called 'miscibility window'. A $T(w^{red})$ section of the 'miscibility window' at a constant mass fraction of carbon dioxide and at constant temperature is shown in Fig. 6a. Here also barotropy occurs at temperatures below 323.2 K.

Because the critical lines of the corresponding binaries of this ternary system are very similar additional isothermal measurements at 308.2 K were carried out. This temperature is situated below both temperature minima of the binary critical lines (Fig. 3). The result is shown in Fig. 4c. Like the isobaric prisms the isothermal phase prism exhibits a homogeneous region inside, leading to an isothermal miscibility window. The section through the isothermal phase prism for a constant mass fraction of carbon dioxide of 0.63 is shown in Fig. 6b; here a homogeneous region larger than 67 MPa is found.

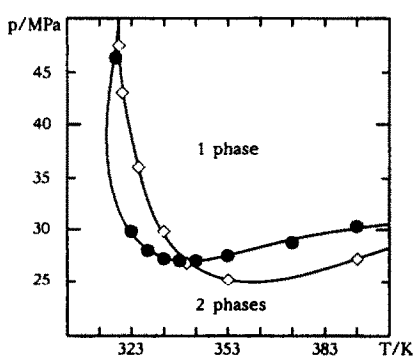


Fig. 3. p, T-projections of the critical curves of the systems CO_2 + 1-dodecanol (Schneider, 1991) (◇) and CO_2 + nonadecane (●).

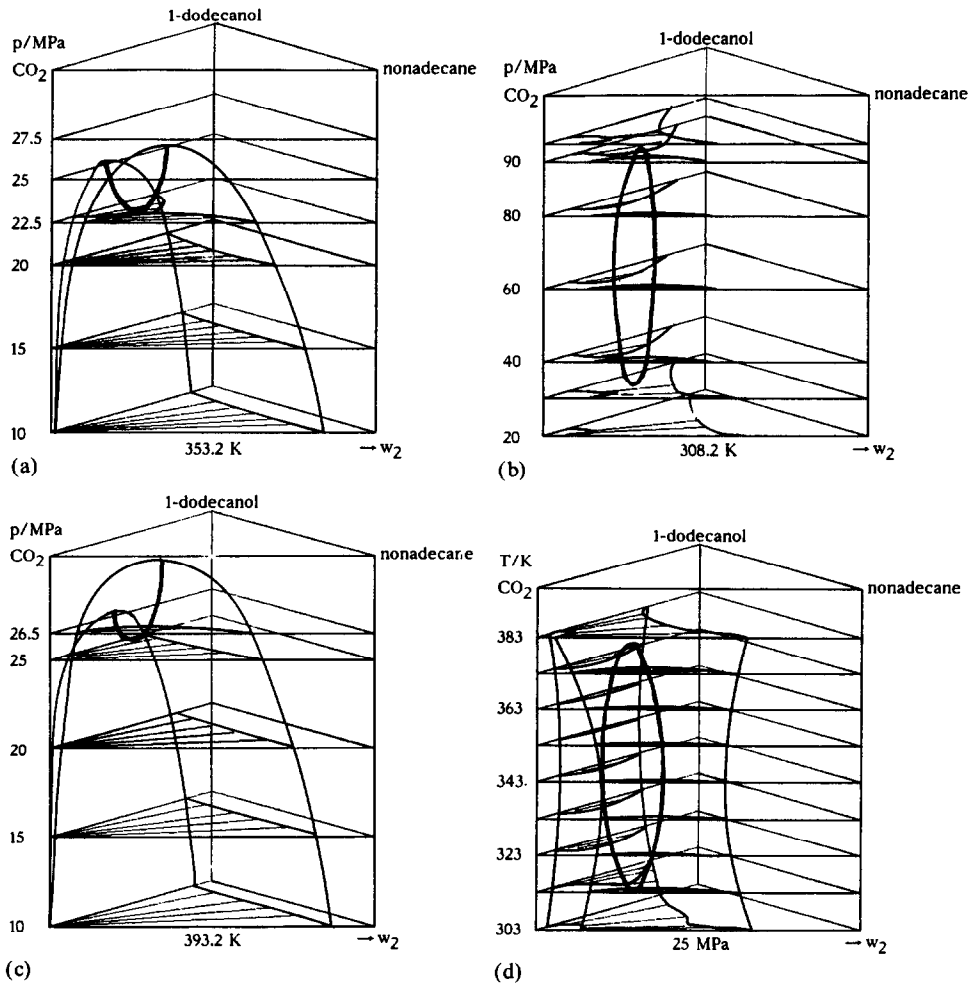


Fig. 4. Ternary phase prisms for the system CO_2 + nonadecane + 1-dodecanol. (a) Isothermal phase prism at 353.2 K. (b) Isothermal phase prism at 393.2 K. (c) Isothermal phase prism at 308.2 K. (d) Isobaric phase prism at 25 MPa.

Carbon dioxide + eicosane + 1-dodecanol

This system was investigated at the same temperatures and pressures as the preceding system. The results are again very similar, and barotropy occurs below 323.2 K. The isothermal and isobaric prisms are similar to those of the preceding system and will not be shown. In the quasibinary projection of the ternary critical curves in Figs. 5a and 5b you can see that the 'cosolvency effect' has significantly decreased and the miscibility window has a distinctly smaller size than that of the system containing nonadecane (Fig. 6a).

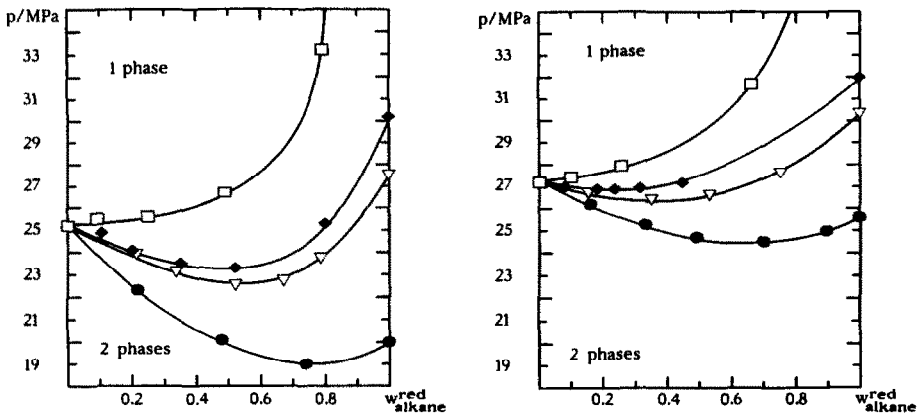


Fig. 5. Ternary critical curves at 353.2 K (a) and 393.2 K (b) of the system CO_2 + hexadecane + 1-dodecanol (393.2; Spee, 1990) (●), CO_2 + nonadecane + 1-dodecanol (▽), CO_2 + eicosane + 1-dodecanol (◆) and CO_2 + tetracosane + 1-dodecanol (□). (w^{red} = carbon dioxide free mass fraction.)

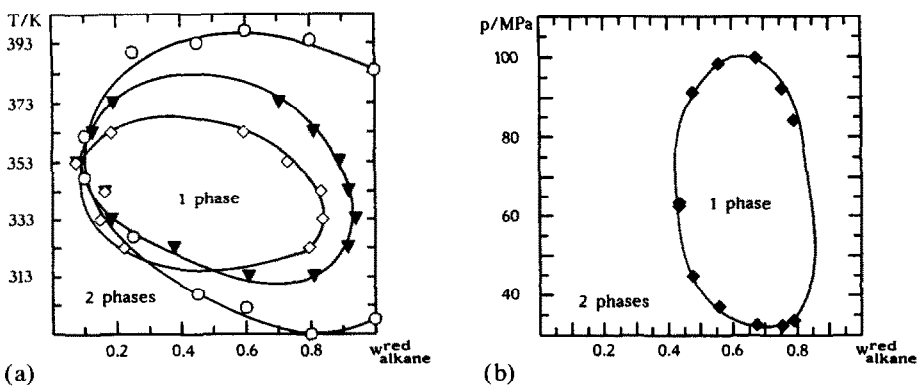


Fig. 6. Miscibility windows in CO_2 + n-alkane + 1-dodecanol systems. (a) Isobaric $T(w^{red})$ sections at 25 MPa and for $w(CO_2) = 0.63$ [n-alkane = hexadecane (Schneider, 1991; Spee, 1990) (○), nonadecane (▼), eicosane (◇)]. (w^{red} = carbon dioxide free mass fraction.) (b) Isothermal $p(w^{red})$ section at 308.2 K and for $w(CO_2) = 0.63$ of the system CO_2 + nonadecane + 1-dodecanol. (w^{red} = carbon dioxide free mass fraction.)

Carbon dioxide + tetracosane + 1-dodecanol

The system CO_2 + tetracosane + 1-dodecanol has been investigated in the same way as the other ternary systems, and again the phase prisms are not shown. In the projection of the isothermal ternary critical curves in Figs. 5a and 5b a pressure minimum on the ternary critical curve no longer exists. It follows consequently that no miscibility window occurs.

DISCUSSION

Figures 5 and 6 demonstrate that here the cosolvency effect and the corresponding miscibility windows decrease with increasing chain length of the alkane but increase with decreasing temperature. With tetracosane in both cases the minimum on the ternary critical curve no longer exists and there is also no miscibility window. Comparing the results with those of Spee (1990) and Schneider (1991) you see that at 25 MPa with hexadecane the miscibility window is not closed on the alkane rich side of the diagram, and that nonadecane is the first alkane studied up to now producing a miscibility window that is closed at both sides.

In the system CO_2 + nonadecane + 1-dodecanol another interesting phenomenon occurs. Because the critical lines of the corresponding binary systems are so similar and the cosolvency effect (as pointed out before) increases with decreasing temperature, an isothermal miscibility window exists at temperatures just below the temperature minima of both binary critical curves.

Combining all results for the system CO_2 + nonadecane + 1-dodecanol and making a projection of the ternary data against a carbon dioxide free mass fraction, a p, T, w^{red}-phase cube is obtained. This phase cube represents the whole phase equilibrium in the ternary system (Fig. 7). It can also be understood as a three-dimensional section of the (in total) four-dimensional ternary phase equilibrium.

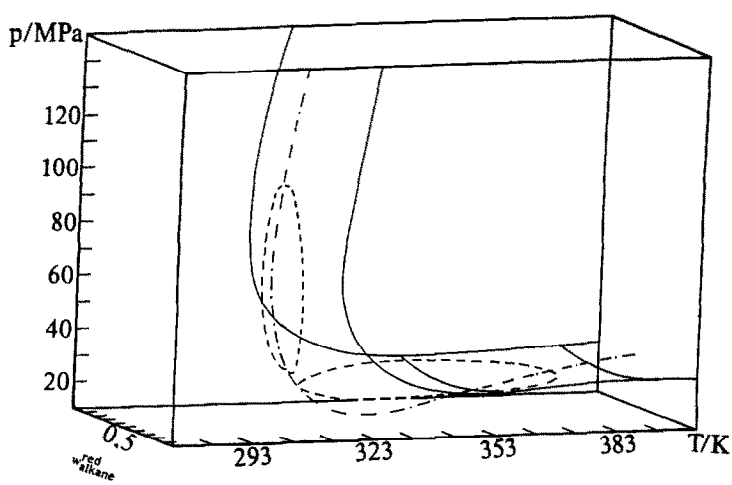


Fig. 7. The CO_2 + nonadecane + 1-dodecanol system in a p, T, w^{red}-phase cube (semiquantitatively; w^{red} = carbon dioxide free mass (fraction). ——— = critical curve (binary, ternary), ·-·-·-·-· = 'global' ternary critical curve, ---- = miscibility window (isobaric, isothermal).

If all minima of the projections of the ternary critical curves and the UCSTs, LCSTs, UCSPs and LCSPs [UCST(P) or LCST(P) = upper or lower critical solution temperature (pressure)] of the miscibility windows are connected respectively, you obtain a 'global' ternary critical curve, which resembles in its p, T-projection the binary critical lines.

It has to be kept in mind, however, that the miscibility windows mentioned above represent sections for a constant mass fraction of carbon dioxide. For the section chosen here [$w(CO_2) \approx 0.63$] the differences between these sections and the miscibility windows such as that shown in Fig. 7 are very small.

CONCLUSION

The present systems were investigated in order to study how the cosolvency effect and the corresponding miscibility windows vary with changing parameters (temperature, pressure and chainlength). It was shown that all parameters have a significant influence on the resulting phase behaviour and that under special conditions new homogeneous regions in ternary systems can be found. For further details and numerical data see Kordikowski (1992).

After submission of the present manuscript the authors were informed about similar effects found in the liquid system carbon dioxide + tetradecane + 1-dodecanol by Patton et al. (1993, in press).

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REFERENCES

Brunner, G., 1983. Selectivity of supercritical compounds and entrainers with respect to model substances. Fluid Phase Equilibria. 10: 289–298.

Deiters, U.K. and Schneider, G.M., 1986. High pressure phase equilibria: Experimental methods. Fluid Phase Equilibria, 29: 145–160.

Hölscher, I.F., Spee, M. and Schneider, G.M., 1989. Fluid phase equilibria of binary and ternary mixtures of CO_2 with hexadecane, 1-dodecanol, 1-hexadecanol and 2-ethoxyethanol at 333.2 K and pressures up to 33 MPa. Fluid Phase Equilibria, 49: 103–113.

Katzenski-Ohling, G. and Schneider, G.M., 1987. Fluid phase equilibria of binary and ternary mixtures of 1-dodecanol with CO_2, $CClF_3$, CHF_3 and N_2 at 392 K and pressures up to 90 MPa. Fluid Phase Equilibria, 34: 273–286.

Konrad, R., 1982. Thermodynamik und Phasenverhalten fluider Dreistoffsysteme bei hohen Drücken. Doctoral Thesis, University of Bochum, 180 pp.

Konrad, R., Swaid, I. and Schneider, G.M., 1983. High-pressure phase studies on fluid mixtures of low-volatile organic substances with supercritical carbon dioxide. Fluid Phase Equilibria, 10: 307–314.

Kordikowski, A., 1972. Fluidphasengleichgewichte ternärer und quaternärer Mischungen schwerflüchtiger organischer Substanzen mit überkritischem Kohlendioxid bei Temperaturen zwischen 298 K and 393 K und Drücken zwischen 10 MPa und 100 MPa. Doctoral Thesis, University of Bochum, 162 pp.

Patton, C.L., Kisler, S.H. and Luks K.D., 1993. Multiphase equilibrium behavior of a mixture of CO_2, 1-decanol and *n*-tetradecane. In E. Kiran and J. Brennecke (Eds.), Supercritical Fluid Engineering Science. ACS Symposium Series No. 514, pp. 55–65.

Schneider, G.M., 1991. High-pressure investigations on fluid systems—A challenge to experiment, theory and application, 8th Rossini Lecture. J. Chem. Thermodynamics, 23: 301–326.

Schneider, G.M., Stahl, E. and Wilke, G. (Eds.), 1980. Extraction with Supercritical Gases, Verlag Chemie, Weinheim, 189 pp.

Spee, M., 1990. Fluidphasengleichgewichte der quaternären Systeme CO_2 + 1-Dodecanol + Hexadecan + 1,8-Octandiol und CO_2 + 1-Dodecanol + Hexadecan + Dotriacontan sowie einiger binärer und ternärer Teilsysteme zwischen 293 K und 413 K bei Drücken bis 100 MPa. Doctoral Thesis, University of Bochum, 120 pp.

Spee, M. and Schneider, G.M., 1991. Fluid phase equilibrium studies on binary and ternary mixtures of carbon dioxide with hexadecane, 1-dodecanol, 1,8-octanediol and dotriacontane at 393.2 K and pressures up to 100 MPa. Fluid Phase Equilibra, 65: 263–274.